

User interface design

Topics covered

- Design issues
- The user interface design process
- User analysis
- User interface prototyping
- Interface evaluation

The user interface

- **User interfaces should be designed to match the skills, experience and expectations of its anticipated users.**
- System users often judge a system by its interface rather than its functionality.
- A poorly designed interface can cause a user to make catastrophic errors.
- Many so-called “user errors” are caused by the fact that user interfaces do not consider the capabilities of real users and their working environment.
- Poor user interface design is the reason why so many software systems are never used.

User Interface Design

- Designing and building software interfaces that are;
 - **Easy to learn**
 - **Easy to use**
 - **Avoid user errors**

Human factors in interface design

- **Limited short-term memory**

- People can instantaneously remember about 7 items of information. If you present more than this, they are more liable to make mistakes.

- **People make mistakes**

- When people make mistakes and systems go wrong, inappropriate alarms and messages can increase stress and hence the likelihood of more mistakes.

- **People are different**

- People have a wide range of physical capabilities. Designers should not just design for their own capabilities.

UI design principles

- **UI design must take account of the needs, experience and capabilities of the system users.**
- **Designers should be aware of people's physical and mental limitations (e.g. limited short-term memory) and should recognise that people make mistakes.**

Design principles -1

- **User familiarity**

- **The interface should be based on user-oriented terms and concepts rather than computer concepts.**

For example, an office system should use concepts such as letters, documents, folders etc. rather than directories, file identifiers, etc. The underlying implementation of the interface in terms of files and data structures should be hidden from the end-user.

- **Consistency**

- **The system should display an appropriate level of consistency.** Commands and menus should have the same format, command punctuation should be similar, etc. Consistent interfaces reduce user learning time. But complete consistency is neither possible nor desirable.

Design principles -2

- **Minimal surprise**

- If a command operates in a known way, the user should be able to predict the operation of similar commands.

- **Recoverability**

- The system should provide some resilience to user errors and allow the user to recover from errors. This might include an undo facility, confirmation of destructive actions etc.

- **User guidance**

- Some user guidance such as help systems, on-line manuals, etc. should be supplied. These should provide different levels of help.

- **User diversity**

- Interaction facilities for different types of user should be supported. For example, some users have seeing difficulties and so larger text should be available or experienced users and inexperienced users etc.

Design issues in UIs

- Two problems must be addressed in interactive systems design:
 - How should information from the user be provided to the computer system?
 - How should information from the computer system be presented to the user?
- User interaction and information presentation may be integrated through a coherent framework such as a **user interface metaphor**.

Interaction styles

- ***Direct manipulation:*** The user interacts directly with objects on the screen. For ex; to delete a file, you may click on an icon representing that file and drag it to a trash can icon.
- ***Menu selection:*** The user selects a command from a list of possibilities.
- ***Form fill-in:*** The user fills in the fields of a form.
- ***Command language:*** The user issues a special command and associated parameters to instruct the system what to do.
- ***Natural language:*** The user issues a command in natural language. The natural language is parsed and translated to system commands.

Interaction styles

Interaction style	Main advantages	Main disadvantages	Application examples
Direct manipulation	Fast and intuitive interaction Easy to learn	May be hard to implement. Only suitable where there is a visual metaphor for tasks and objects.	Video games CAD systems
Menu selection	Avoids user error Little typing required	Slow for experienced users. Can become complex if many menu options.	Most general-purpose systems
Form fill-in	Simple data entry Easy to learn Checkable	Takes up a lot of screen space. Causes problems where user options do not match the form fields.	Stock control, Personal loan processing
Command language	Powerful and flexible	Hard to learn. Poor error management.	Operating systems, Command and control systems
Natural language	Accessible to casual users Easily extended	Requires more typing. Natural language understanding systems are unreliable.	Information retrieval systems

Web-based interfaces

- Since web-based systems are usually designed for casual users, most of them use form-based interfaces.
- Form field can be menus, free text input, radio buttons, etc.
- In the LIBSYS example, users make a choice of where to search from a menu and type the search phrase into a free text field.(document search and document request)

LIBSYS search form

LIBSYS: Search

Choose collection

All



Keyword or phrase

Search using

Title



Adjacent words



Yes



No

Search

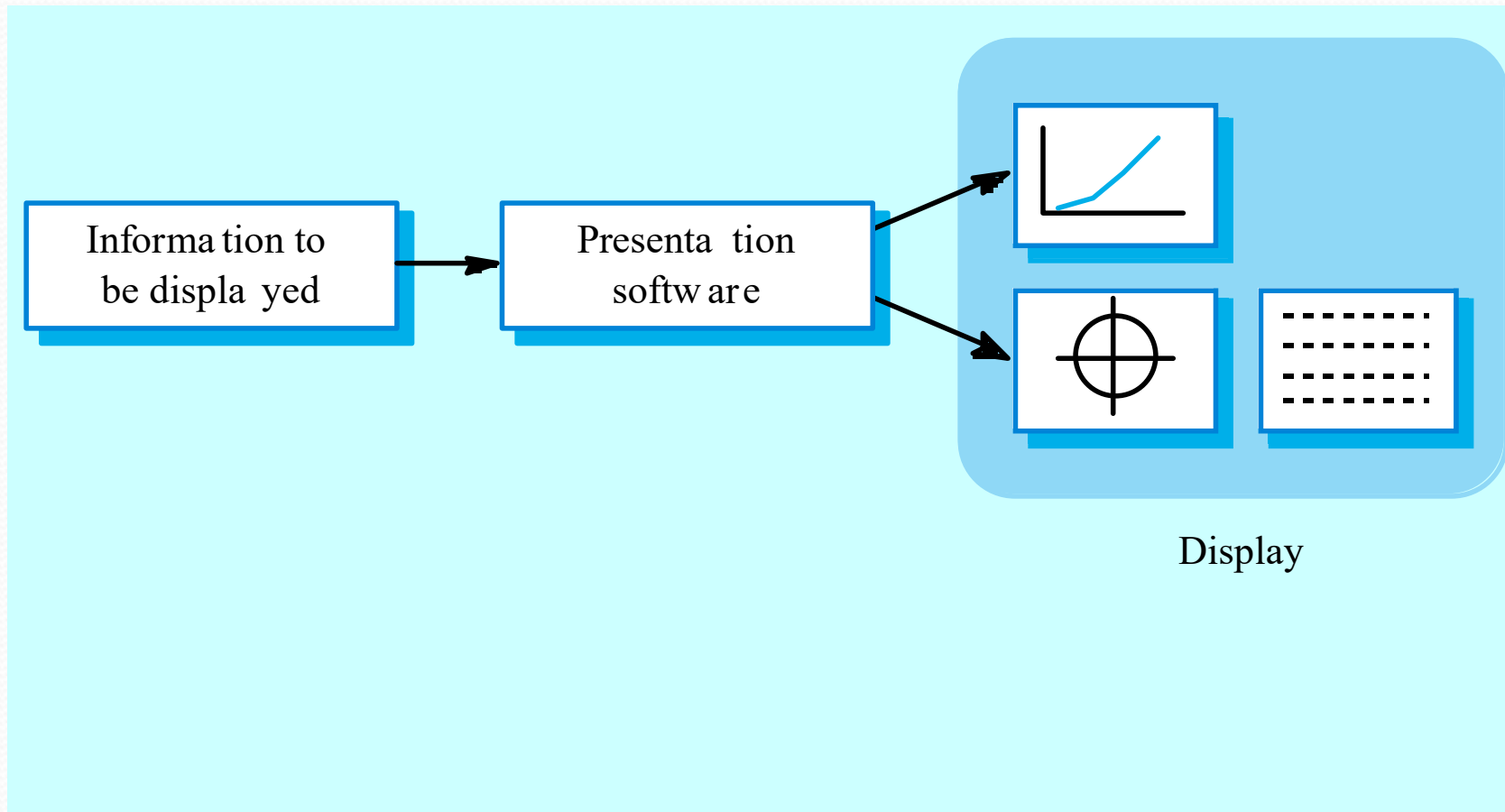
Reset

Cancel

Information presentation

- **Information presentation is concerned with presenting system information to system users.**
- **The information may be presented directly (e.g. text in a word processor) or may be transformed in some way for presentation (e.g. in some graphical form).**
- **The Model-View-Controller approach is a way of supporting multiple presentations of data.**

Information presentation



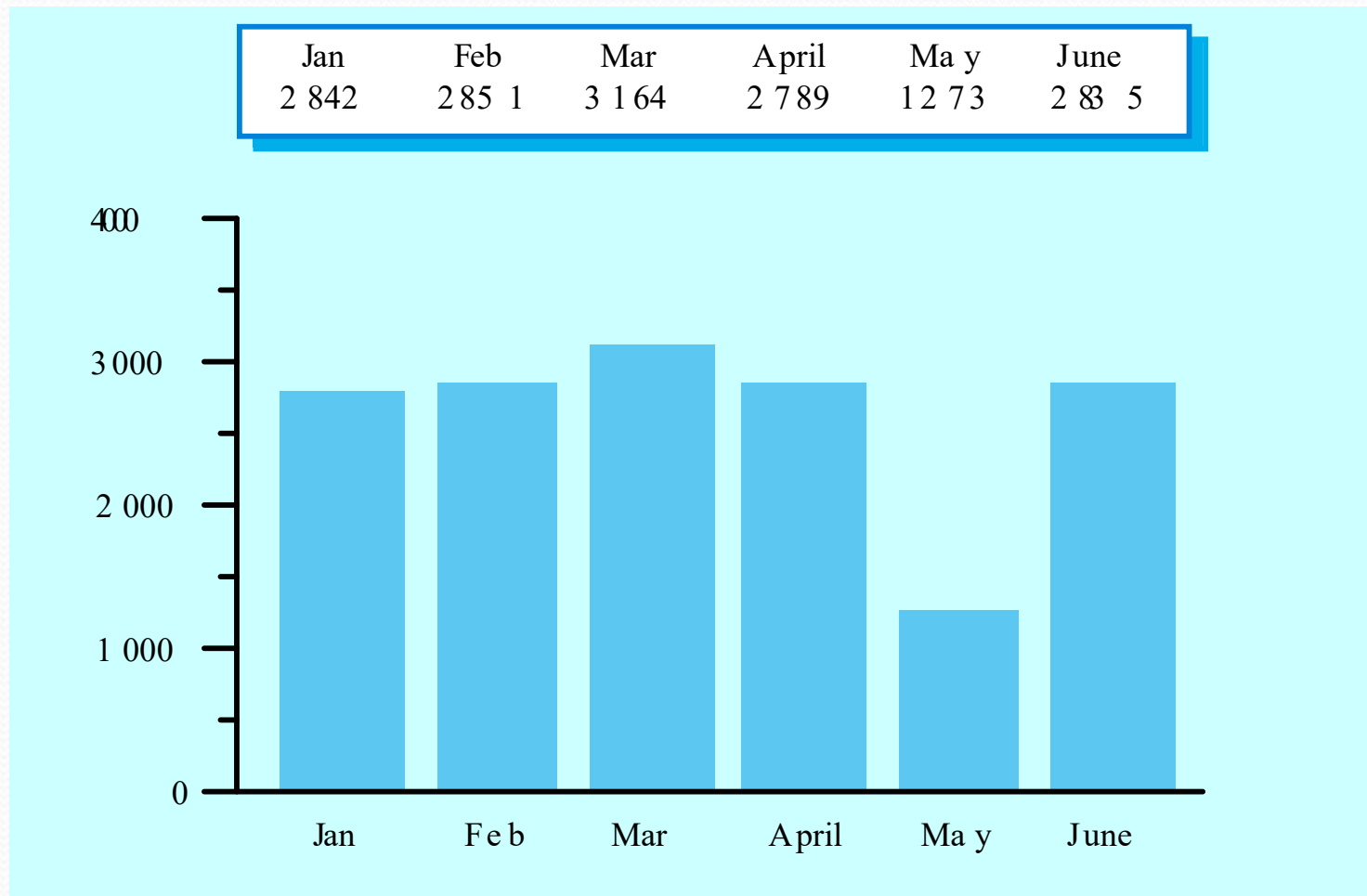
Information presentation

- **Static information**
 - Initialised at the beginning of a session. It does not change during the session.
 - May be either numeric or textual.
- **Dynamic information**
 - Changes during a session and the changes must be communicated to the system user.
 - May be either numeric or textual.

Information display factors

- Is the user interested in precise information or data relationships?
- How quickly do information values change? Must the change be indicated immediately?
- Must the user take some action in response to a change?
- Is there a direct manipulation interface?
- Is the information textual or numeric? Are relative values important?

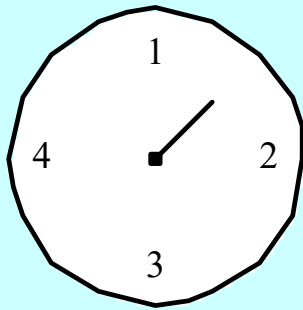
Alternative information presentations



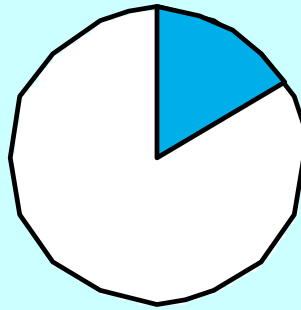
Analogue or digital presentation?

- Digital presentation
 - Compact - takes up little screen space;
 - Precise values can be communicated.
- Analogue presentation
 - Easier to get an 'at a glance' impression of a value;
 - Possible to show relative values;
 - Easier to see exceptional data values.

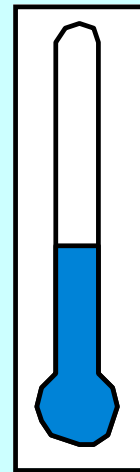
Presentation methods



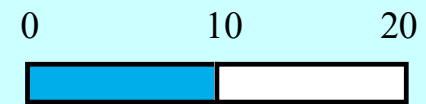
Dial with needle



Pie chart

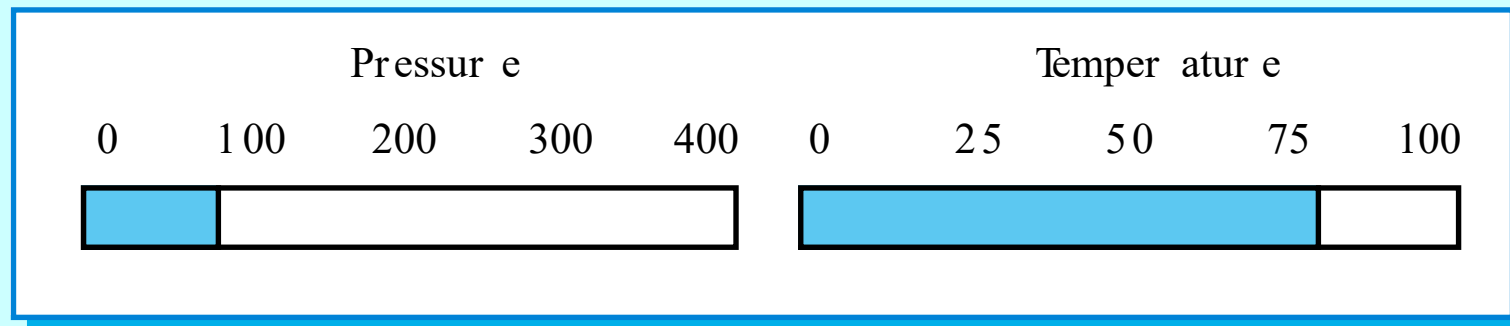


Thermometer



Horizontal bar

Displaying relative values



Data visualisation

- Concerned with techniques for displaying large amounts of information.
- Visualisation can reveal relationships between entities and trends in the data.
- Possible data visualisations are:
 - Weather information collected from a number of sources;
 - The state of a telephone network as a linked set of nodes;
 - Chemical plant visualised by showing pressures and temperatures in a linked set of tanks and pipes;
 - A model of a molecule displayed in 3 dimensions;

Colour displays

- Colour adds an extra dimension to an interface and can help the user understand complex information structures.
- Colour can be used to highlight exceptional events.
- An example for a common mistake in the use of colour in interface design include the **over-use of colour** in the display.

Colour use guidelines

- Limit the number of colours used and be conservative in their use.
- Use colour change to show a change in system status.
- Use colour coding in a thoughtful and consistent way.

Error messages

- *Error message design is critically important.*

Poor error messages can mean that a user rejects rather than accepts a system.

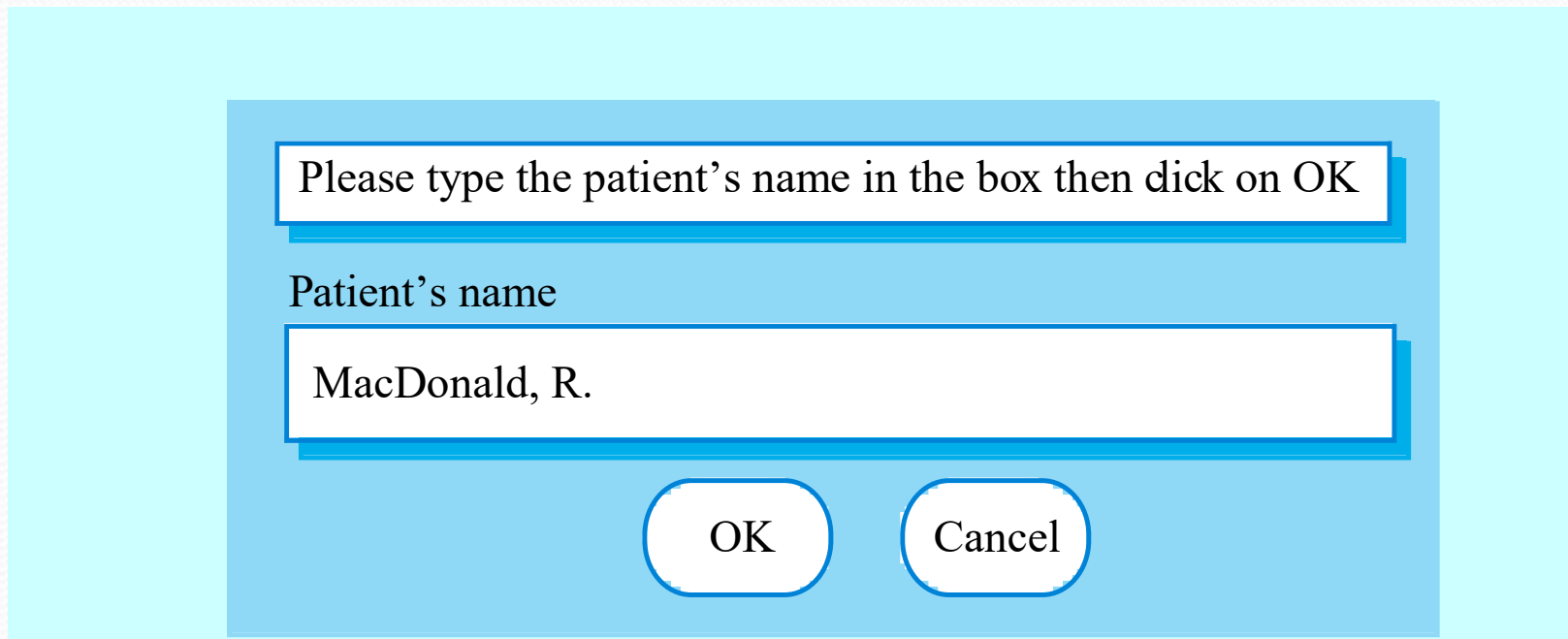
- **Messages should be polite, concise, consistent and constructive.**
- **The background and experience of users should be the determining factor in message design.**

Design factors in message wording

Factor	Description
Context	Wherever possible, the messages generated by the system should reflect the current user context. As far as is possible, the system should be aware of what the user is doing and should generate messages that are relevant to their current activity.
Experience	As users become familiar with a system they become irritated by long, 'meaningful' messages. However, beginners find it difficult to understand short terse statements of a problem. You should provide both types of message and allow the user to control message conciseness.
Skill level	Messages should be tailored to the user's skills as well as their experience. Messages for the different classes of user may be expressed in different ways depending on the terminology that is familiar to the reader.
Style	Messages should be positive rather than negative. They should use the active rather than the passive mode of address. They should never be insulting or try to be funny.
Culture	Wherever possible, the designer of messages should be familiar with the culture of the country where the system is sold. There are distinct cultural differences between Europe, Asia and America. A suitable message for one culture might be unacceptable in another.

User error

- Assume that a nurse misspells the name of a patient whose records he is trying to retrieve.



Please type the patient's name in the box then click on OK

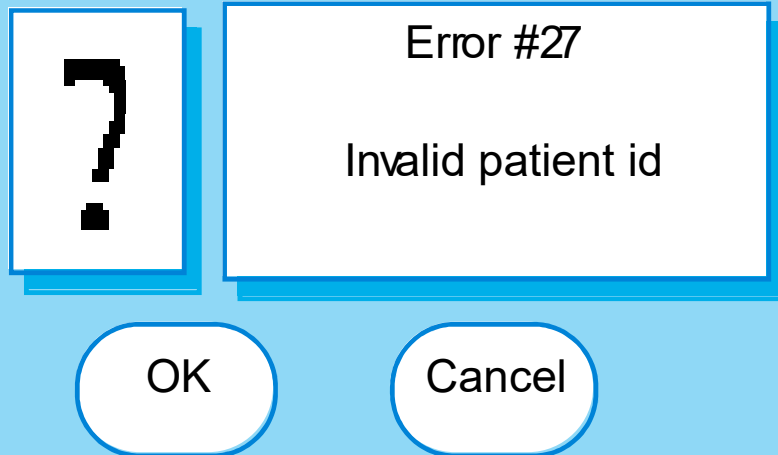
Patient's name

MacDonald, R.

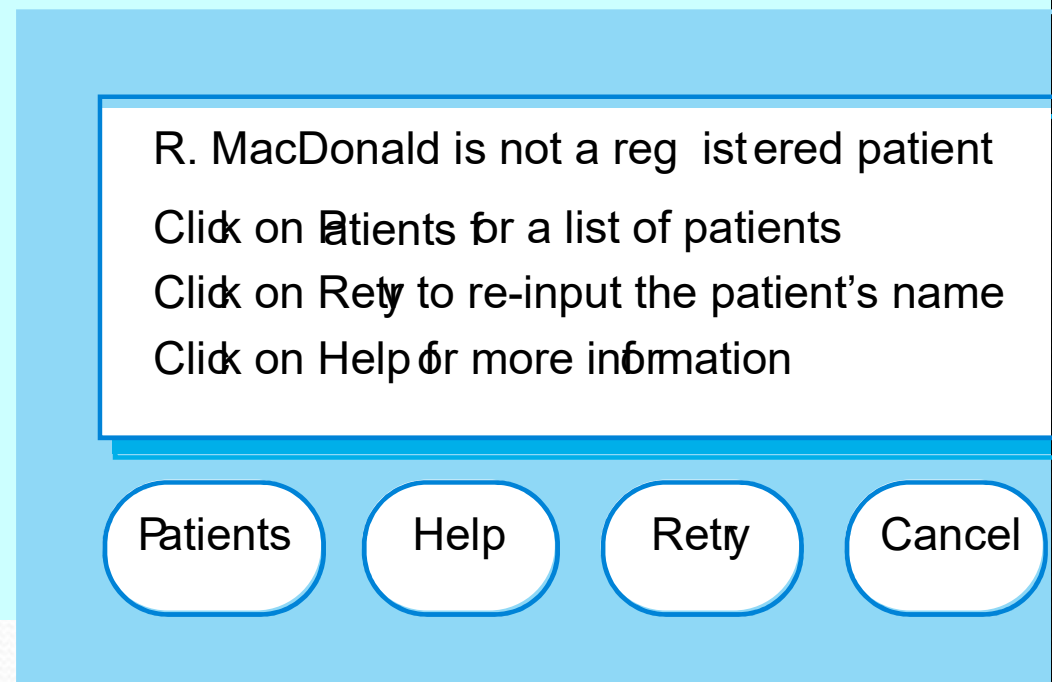
OK Cancel

Good and bad message design

System-oriented error message



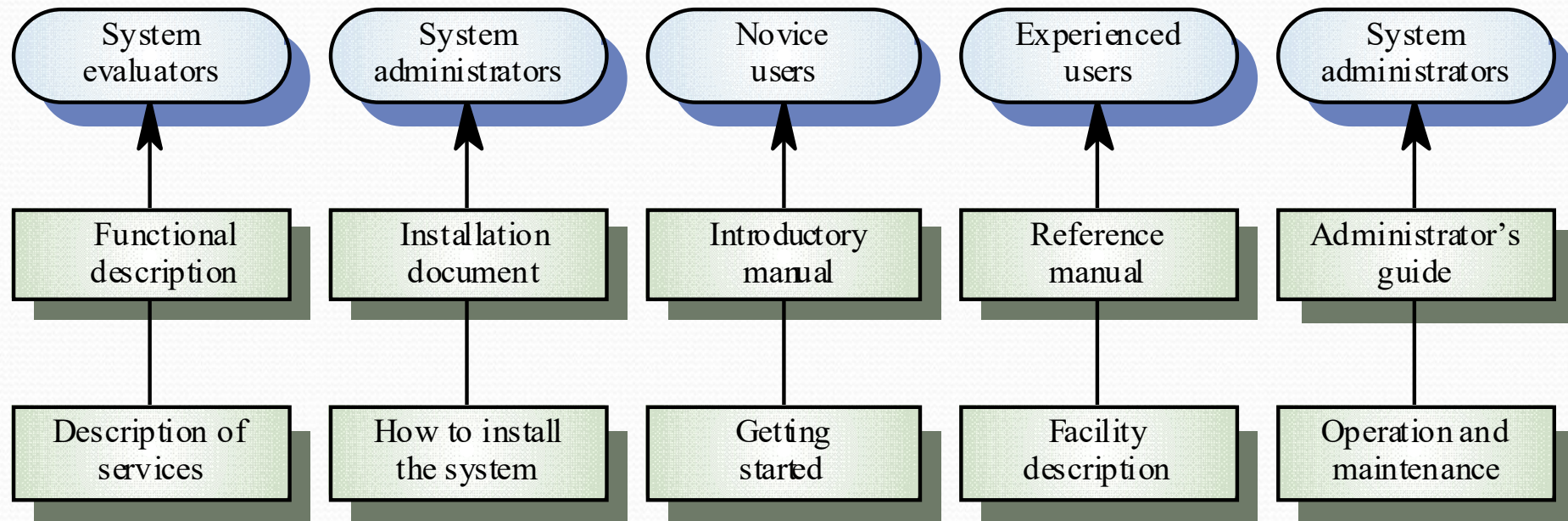
User-oriented error message



User documentation

- As well as on-line information, paper documentation can be supplied with a system
- Documentation should be designed for a range of users from inexperienced to experienced
- As well as manuals, other easy-to-use documentation such as a quick reference card may be provided

User document types



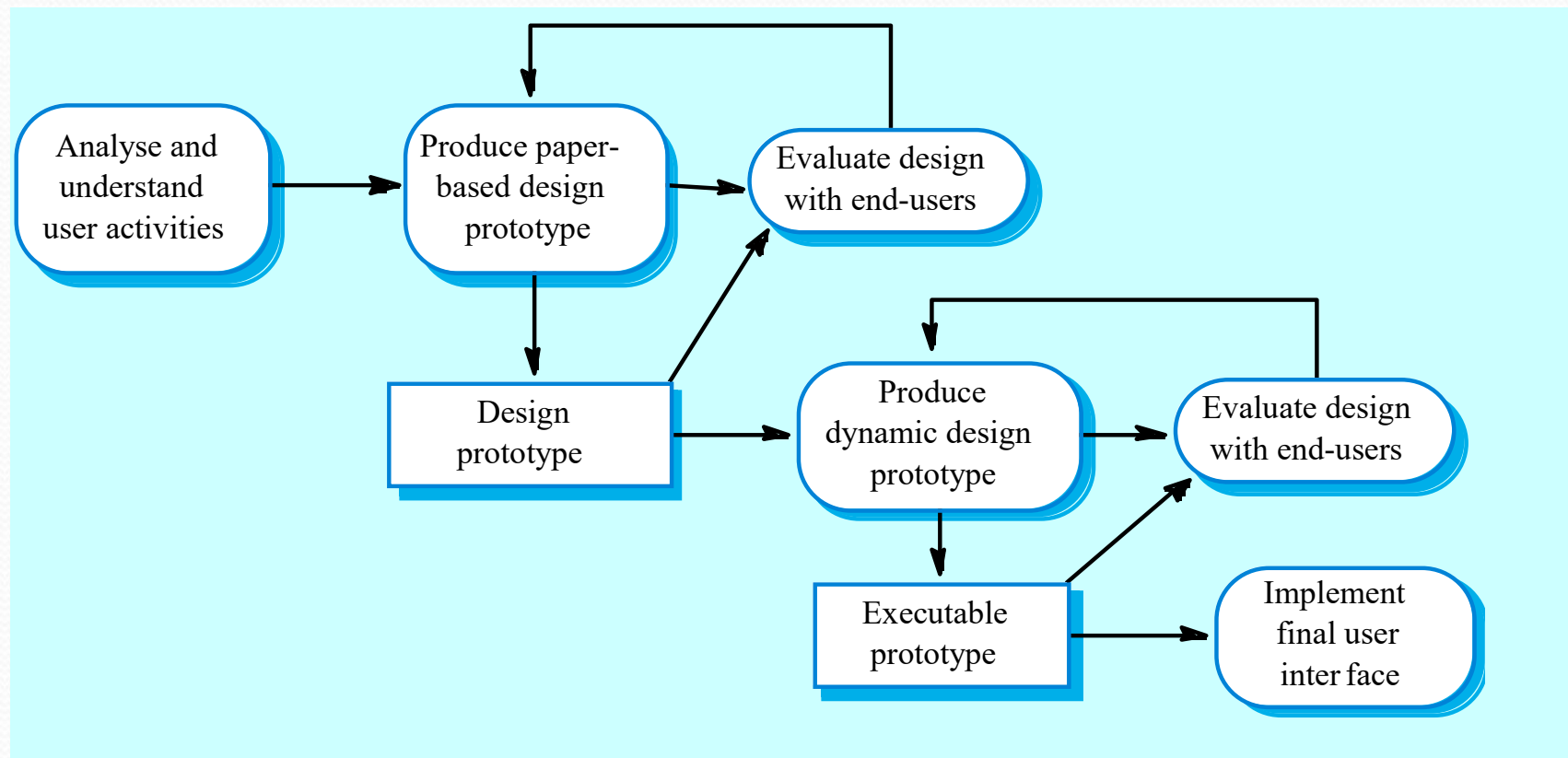
Document types

- **Functional description:** Brief description of what the system can do
- **Introductory manual:** Presents an informal introduction to the system
- **System reference manual:** Describes all system facilities in detail
- **System installation manual:** Describes how to install the system
- **System administrator's manual:** Describes how to manage the system when it is in use

The UI design process

- UI design is an iterative process between users and designers.
- The 3 core activities in this process are:
 - **User analysis.** Understand what the users will do with the system;
 - **System prototyping.** Develop a series of prototypes for experiment;
 - **Interface evaluation.** Experiment with these prototypes with users.

The design process



User analysis

- If you don't understand what the users want to do with a system, you have no realistic prospect of designing an effective interface.
- User analyses have to be described in terms that users and other designers can understand.
- Scenarios where you describe typical episodes of use, are one way of describing these analyses.

Analysis techniques

- **Task analysis**
 - Models the steps involved in completing a task.
- **Interviewing and questionnaires**
 - Asks the users about the work they do.
- **Ethnography**
 - Observes the user at work.

Interviewing

- Design semi-structured interviews based on open-ended questions.
- Users can then provide information that they think is essential; not just information that you have thought of collecting.
- Group interviews and allow users to discuss with each other what they do.

Ethnography

- Involves an external observer watching users at work and questioning them in an unscripted way about their work.
- Valuable because many user tasks are intuitive and they find these very difficult to describe and explain.
- Also helps understand the role of social and organisational influences on work.

User interface prototyping

- The aim of prototyping is to allow users to gain direct experience with the interface.
- Without such direct experience, it is impossible to judge the usability of an interface.
- Prototyping may be a two-stage process:
 - Early in the process, paper prototypes may be used;
 - The design is then refined and increasingly sophisticated automated prototypes are then developed.

Paper prototyping

- Work through scenarios using sketches of the interface.
- Use a storyboard to present a series of interactions with the system.
- Paper prototyping is an effective way of getting user reactions to a design proposal.

User interface evaluation

- **Some evaluation of a user interface design should be carried out to assess its suitability.**
- **Full scale evaluation is very expensive and impractical for most systems.**
- **Ideally, an interface should be evaluated against a usability specification. However, it is rare for such specifications to be produced.**

Usability attributes

Attribute	Description
Learnability	How long does it take a new user to become productive with the system?
Speed of operation	How well does the system response match the user's work practice?
Robustness	How tolerant is the system of user error?
Recoverability	How good is the system at recovering from user errors?
Adaptability	How closely is the system tied to a single model of work?

Simple evaluation techniques

- Questionnaires for user feedback.
- Video recording of system use and subsequent tape evaluation.
- Instrumentation of code to collect information about facility use and user errors.
- The provision of code in the software to collect on-line user feedback.